AI23521 BUILD AND DEPLOY FOR MACHINE LEARNING APPLICATION

PROJECT: SMS SPAM DETECTION USING MACHINE LEARNING

AIM:

To develop a machine learning application that classifies SMS messages as **spam** or **ham (legitimate)** using text preprocessing, TF-IDF feature extraction, and a Multinomial Naive Bayes classifier.

ABSTRACT:

SMS spam messages may contain unwanted advertisements, fraudulent offers, or suspicious links. This project demonstrates a text-classification pipeline that learns patterns from labelled messages and predicts the category of a new message. Text is converted into numerical TF-IDF features and classified using Multinomial Naive Bayes. Python and scikit-learn are used. A larger labelled SMS dataset can be used to extend the demonstration.

PROPOSED SYSTEM:

The system accepts a message, transforms it into TF-IDF features, and passes the features to a trained classifier. The predicted label is displayed as SPAM or HAM. A small built-in example dataset allows the program to run without downloading external data. Since this dataset is intentionally small, its accuracy is not representative of real-world performance.

ALGORITHM:

1. Import the required Python libraries.
2. Prepare labelled SMS messages belonging to ham and spam classes.
3. Split the messages and labels into training and testing sets.
4. Convert text into numerical features using TF-IDF Vectorizer.
5. Train a Multinomial Naive Bayes classifier on the training features.
6. Predict test labels and calculate accuracy.
7. Transform a new SMS using the same vectorizer and predict its category.
8. Display the predicted category.

CODE:

```
# SMS Spam Detection using TF-IDF and Naive Bayes
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score

messages = [
    "Hey, are we meeting for lunch today",
    "Please call me when you reach home",
    "Your appointment is confirmed for tomorrow",
    "Can you send me the class notes",
    "Congratulations you have won a free prize",
    "Claim your cash reward now by clicking this link",
    "URGENT you have been selected for a free gift",
    "Win a voucher now call this number",
    "Reminder: your project meeting starts at ten",
    "I will be late to class today",
    "You won a lottery claim your reward now",
    "Free entry in a weekly prize draw"
]

labels = [
    "ham", "ham", "ham", "ham", "spam", "spam",
    "spam", "spam", "ham", "ham", "spam", "spam"
]

X_train, X_test, y_train, y_test = train_test_split(
    messages, labels, test_size=0.25, random_state=42, stratify=labels
)

model = Pipeline([
    ("tfidf", TfidfVectorizer(lowercase=True, stop_words="english")),
    ("classifier", MultinomialNB())
])

model.fit(X_train, y_train)
predictions = model.predict(X_test)
print("Test accuracy:", round(accuracy_score(y_test, predictions), 2))

new_message = "Congratulations! Claim your free cash prize now"
prediction = model.predict([new_message])[0]
print("Message:", new_message)
print("Predicted category:", prediction.upper())
```

SAMPLE OUTPUT:

The actual accuracy is calculated when the program runs. Sample output format:

```
Test accuracy: <calculated value>\nMessage: Congratulations! Claim your free cash prize now\nPredicted cat
```

Note: The test accuracy depends on the selected split and dataset. The small example dataset is for demonstrating the workflow only.

RESULT:

The SMS Spam Detection application was implemented using TF-IDF feature extraction and a Multinomial Naive Bayes classifier. The system trains on labelled text messages and predicts whether a new message belongs to the spam or ham category. Thus, a basic end-to-end machine learning workflow was demonstrated using Python.